

POWER SUPPLY FOR CLOCK

This is regarding the article on Power Supply for a Clock Using Supercapacitor and Small Solar Panels, which was published in February issue. Is this economical enough? A supercapacitor costs from ₹200 up to ₹7,500 and a solar panel costs from ₹100 up to ₹3,500. So, is this the right alternative for lithium-ion or alkaline batteries? Or, are there suitable means to increase their overall storage capacity so that they can replace the conventional primary or secondary batteries? Although a supercapacitor holds charge in the range of tens of Farads, their delivery of power over a long period is still below the actual field requirements.

Chithraa

EFY. The author Dr Vijay Deshpande replies:

“This is a good observation. The demand for energy storage devices is increasing exponentially. Following are broad guidelines through which we can try to meet the demand.

1. Explore all types of storage devices.
2. Try to take advantage of their unique features. For example, supercapacitors can be charged and discharged 100,000 times. So, they have three times more life than a lithium-ion battery—even after charging and discharging several times in a day. Also, supercapacitors are environment friendly.

3. The cost issue can be tackled by increasing demand and production. For that, we have to identify applications that are low lying fruits. This way, we can generate initial demand, which will result in reduced price. In turn, it will open up newer applications and improved capacitors.

The clock power supply circuit was designed with above points in mind. Even though it is a very small power application, it has the potential to eliminate tens of millions of AA and

AAA batteries, thereby reducing their hazardous waste. Use of lithium-ion batteries for stationary applications is a waste of precious resource.

As far as solar panels are concerned, they have a long life of over 25 years. Problem is the initial cost. That is the reason the government is offering subsidies.”

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SMART WATCH

This is regarding the DIY article titled Wi-Fi Analysing Smart Watch published in March issue. How much current to ESP32 or ESP8266 GPIO pins can you provide? Can I directly connect the micro-sized vibration motor to the ESP8266?

Bidrohini

EFY. The maximum output current from one GPIO pin of ESP8266 is 12mA. In the case of ESP32, it is 40mA. Connecting a micro-sized motor directly to ESP8266 is a bad idea. Use a motor driver IC.

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BATTERYLESS REMOTE PROJECT

What is the voltage range of LTC3588 and how much does the IC cost?

Gayathri

EFY. The piezoelectric energy harvester module LTC3588 can harvest energy from a selectable range of 1 to 3.3V. Its cost is around one thousand Indian rupees.

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OLED USE WITH ARDUINO

Which Arduino is used for making the smart glasses using an OLED display?

Noxious

EFY. You can use any version of Arduino. The author used Arduino nano

because of its small size in the project published in November issue.

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HELP FOR ELECTRONIC PROJECT

How can we contact the authors for some help regarding their articles or projects published in Electronics For You as well as electronicsforu.com?

Aryan Surve

EFY. You can email your query to onedit@efy.in or discuss with experts using the EFY forum <https://www.electronicsforu.com/forums>

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SUBMISSION OF PROJECT

What are the rules for submitting complete DIY project details for publication in Electronics For You?

Chithraa

EFY. Email your project details to onedit@efy.in or directly submit on <https://www.electronicsforu.com/submit-your-project>. The project will be published after review by EFY's editorial team and their approval.

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ERRATA

On page No. 91 in EFY January issue under the heading Circuit and Working in DIY article Temperature Controlled Switch, the op-amp is mentioned as LM35 (IC3), while in Fig. 2 on the same page the op-amp IC is mentioned as LM358 (IC3A).

Sachin Goel

EFY. We thank Sachin for detecting this error and sharing with everyone! Please refer the circuit diagram for the correct nomenclature.

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